

Programme	BSc. Computer Science				
Course Title	Automation and Robotics				
Type of Course	Vocational Minor				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	No pre-requisites required				
Course Summary	This course provides a comprehensive overview of automation which includes their production systems, elements, automation functions and usage of discrete and continuous control system. The course also explores the fundamentals of robotics, including anatomy, process control and how these functions could be improved by the integration of Artificial Intelligence.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the production systems and automation, enabling them to analyse, optimize and evaluate the different levels of automation.	U	C	Instructor- created exams / Quiz/Assignment/ Seminar
CO2	Able to recognize the difference between the process industries, manufacturing industries, continuous and discrete control system.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar
CO3	Proficiency in understanding the various forms of process control which includes the direct digital control, programmable logic control, distributable control systems etc.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar
CO4	Familiarize with the various hardware components used for automation and process control such as sensors, actuators analog-digital converters etc.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar

CO5	Understand the present developments in the field of automation and robotics and how integrating artificial intelligence can contribute to the future of these systems.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar
CO6	Represent various problems using algorithmic approaches and enhance problem-solving skills by visualizing solutions through the utilization of software tools.	U, Ap	C, P	Practical Assignment / Observation of Practical Skills
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Contents	Hrs (45+30)	Mark
I	Introduction to Automation		11	15
	1	Production systems - Facilities, Manufacturing support systems	2	
	2	Automation in production systems – Automated manufacturing system, Computerized manufacturing support systems, Reasons for automating	3	
	3	Manual labour in production systems	1	
	4	Elements of automation - power to accomplish the process, Program of instructions, control system	3	
	5	Advanced automation functions – safety monitoring, maintenance and repair diagnostics, error detection and recovery	1	
	6	Levels of automation	1	
II	Control Systems		13	15
	7	Process industries versus Discrete manufacturing industries, Continuous versus Discrete control	1	
	8	Continuous control system	3	
	9	Discrete control system	1	
	10	Computer process control, Control requirements, Capabilities of computer control	2	

	11	Forms of computer process control - Computer process monitoring, Direct digital control, Computer numerical control and robotics, Programmable logic controllers, Supervisory control and data acquisition, Distributed control systems	3	
	12	Hardware for automation and process control (Concept only) - Sensors, Actuators, Analog to Digital converters Digital to Analog converters, Input/output devices for discrete data.	3	
III	Industrial Robotics		15	25
	13	Robot anatomy – Joints and links, Common robot configurations, Joint drive systems, Sensors in robotics	4	
	14	Robot control systems – Limited sequence control, Playback with point-to-point control, Playback with continuous path control, Intelligent control	2	
	15	End effectors – Grippers, Tools	1	
	16	Robot Programming – Lead through programming, Powered lead through, Motion programming, Advantages and disadvantages	2	
	17	Discrete process control – logic control, sequence control	4	
	18	Programmable Logic Controllers, Components of PLC	2	
	Automation and Robotics: Present and Future		6	
IV	19	Machine Intelligence, Computer and Robotics	1	15
	20	Flexible automation vs Robotics technology	1	
	21	Artificial Intelligence and Automated Manufacturing, AI and Robotics	2	
	22	Robotics in India, Future of Robotics	2	
	Lab Activities		30	
V	1	Set up a simulation of a production system using any software tools.	28	
	2	Utilise online simulation tools and platforms that allow students to simulate robot control.		
	3	Utilise online simulation tools and platforms that allow students to simulate automation systems.		
	4	Assign online projects or challenges that require participants to design, program, or simulate automation systems and robotic applications.		

	5	Explore any online virtual reality (VR) applications that simulate manufacturing environments, robotic operations, and automation scenarios.		
	6	Analyze publicly available datasets on platforms like Kaggle, UCI Machine Learning Repository, or Data.gov.		
	7	Experiment with virtual robotics simulations using platforms like V-REP (Virtual Robot Experimentation Platform) or Gazebo.		
	8	Designing and building a simple chatbot using no-code platforms like ChatGPT or Google's Dialogflow.		
	9	Allow students to customize their chatbots by defining conversational flows.		
	10	Provide Programmable Logic Controllers (PLCs) and challenge them to program various control sequences.		
	11	Host a discussion session on the intersection of Artificial Intelligence (AI) and Robotics in automated manufacturing.	2	

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	-	-	-	-						
CO 2	2	2	-	-	2	-						
CO 3	2	2	-	-	2	-						
CO 4	2	2	-	-	2	-						
CO 5	1	-	-	-	-	1						
CO 6	-	-	2	2	-	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓

References:

1. Mikell P. Groover, "Automation, Production Systems and Computer Integrated Manufacturing", 4th edition, Pearson Education, 2017.
2. S.R. Deb, S. Deb " Robotics Technology and flexible automation," Tata McGraw-Hill Education, 2017.
- 3 Mikell P. Groover, " "Industrial Robots - Technology, Programming and Applications", McGraw-Hill Education, 2017